

Lab 1 — Data Lake vs Data Warehouse vs Lakehouse

Business Scenario

"BrightCart," an online retailer, has two teams:

- The Finance team needs a dashboard every morning showing yesterday's revenue by region. They need it fast, accurate, and always in the same table format.
- The Data Science team wants to keep every raw file that ever comes in — order logs, app clickstreams, product images — even messy or incomplete ones, because they might use it for future ML models.

Think About It

- (a) Which storage approach fits Finance's need — lake, warehouse, or lakehouse? Why?
- (b) Which fits the Data Science team's need? Why?

A – warehouse because they need accurate and same table format which is structured so data warehouses will have structured data

B – Data lake because it is a raw data coming and it comes with no consistency so

Multiple Choice Questions

MCQ 1.1 — A data warehouse is best described as:

- ☐ A) Cheap storage for any file type, queried later
- ☐ B) A structured repository with a fixed schema, optimized for fast business reporting
- ☐ C) A folder of random files with no organization
- ☐ D) Something used only for videos and images

Answer – B

MCQ 1.2 — Which statement is TRUE about a lakehouse?

- ☐ A) It's just a data lake renamed for marketing purposes
- ☐ B) It combines the low-cost flexible storage of a lake with the reliability and structure of a warehouse
- ☐ C) It can only be used for reporting, never for raw data
- ☐ D) It removes the need for any data cleaning

Answer – B

MCQ 1.3 — Scenario: A hospital wants to store years of unstructured MRI scan images cheaply, and separately run structured SQL reports on patient billing. What's the most sensible setup?

- ☐ A) One data warehouse for everything
- ☐ B) A data lake for MRI images, a data warehouse (or lakehouse) for billing reports
- ☐ C) A single spreadsheet
- ☐ D) No storage needed — just recompute every time

Answer - B

MCQ 1.4 — Scenario: A streaming platform wants one place where engineers can run SQL directly on raw viewing logs, get safe/reliable updates, but still keep storage costs low like a lake. What are they really asking for?

- ☐ A) A traditional on-premise data warehouse
- ☐ B) A lakehouse
- ☐ C) A single Excel workbook
- ☐ D) A backup tape system

Answer – B

MCQ 1.5 — Which of these is a genuine downside of a traditional data warehouse compared to a lake?

- ☐ A) It can't run SQL queries
- ☐ B) It's typically more expensive to scale and less flexible with unstructured/semi-structured data
- ☐ C) It never supports business reporting
- ☐ D) It can only store images

Answer – B

Lab 2 — Schema-on-Read vs Schema-on-Write

Business Scenario

"BrightCart" receives clickstream data from its app. Some events include a "page_name" field, some don't (older app versions didn't record it). The Data Science team wants to store this data immediately without rejecting anything. Meanwhile, Finance's revenue reports must always have exactly the same columns, correctly typed, or the dashboard breaks.

Think About It

- Which team benefits from "decide the structure later" (schema-on-read)?
- Which team needs "structure decided before storage" (schema-on-write)?
- What's the risk of using schema-on-read for Finance's revenue report?

A – Data science team needs schema on read because they want to store all the data immediately without rejecting anything so that later they can apply structure by analysing

B – Finance revenue team need schema on write because they want fixed consistent data

C – If Finance team used schema on read then due to missing field like page_name they may have dashboard breaks due to inconsistent or missing data

Multiple Choice Questions

MCQ 2.1 — Schema-on-read means:

- ☐ **A)** The structure/format is enforced before storing the data
- ☐ **B)** The structure is only figured out when someone actually queries/reads the data
- ☐ **C)** Data can never be read again once written
- ☐ **D)** Only applies to video files

Answer- B

MCQ 2.2 — A downside of schema-on-read is:

- ☐ **A)** It makes writing data slower
- ☐ **B)** Because nothing is validated at ingestion, messy or missing fields can cause problems later when someone tries to use the data
- ☐ **C)** It only works with spreadsheets
- ☐ **D)** It requires a data warehouse license

Answer – B

MCQ 2.3 — Scenario: A bank's transaction system must reject any incoming record missing an account number, before it's ever stored. This is an example of:

- ☐ **A)** Schema-on-read
- ☐ **B)** Schema-on-write
- ☐ **C)** No schema at all
- ☐ **D)** Unstructured storage

Answer – B

MCQ 2.4 — Scenario: A social media company ingests millions of posts per day in constantly-changing formats (new fields added by app updates every month). Enforcing a strict schema at write time would mean:

- ☐ **A)** Nothing changes, it's the same either way
- ☐ **B)** Frequent pipeline failures or rejected records every time the format changes slightly
- ☐ **C)** Faster ingestion with zero downsides
- ☐ **D)** The posts would automatically fix themselves

Answer - B

MCQ 2.5 — Which best describes the general trade-off between schema-on-read and schema-on-write?

- ☐ **A)** Schema-on-read = flexible ingestion but more work later; schema-on-write = more upfront effort but more reliable/consistent data later
- ☐ **B)** They are exactly the same, just different names
- ☐ **C)** Schema-on-write is always better in every situation
- ☐ **D)** Schema-on-read only works for numeric data

Answer- A

Lab 3 — The Bronze (Raw) Zone

Business Scenario

BrightCart's engineering team ingests raw order data every day exactly as it arrives from the checkout system — no cleaning, no filtering — into a "Bronze" folder, organized by the date it arrived.

Think About It

- Why would a company deliberately keep messy, unprocessed data instead of cleaning it immediately?
- What happens if a bug is found in tomorrow's cleaning logic — how does having Bronze data help?

- A- The team keeps messy unprocessed data because for auditability, reprocessing (you can always come back and re run if any bug found in cleaned data or cleaning logic) and to keep the source of truth for downstream transformations and since its a raw zone layer it is a cheap storage so they keep raw data as it is in the bronze layer
- B- Since it is a Bronze layer you can always come back and reprocess and re run the cleaning logic and pipeline and since it is saved in date format it you can just go to that particular day data and re run

Multiple Choice Questions

MCQ 3.1 — The Bronze (raw) zone typically contains:

- ☐ **A)** Only the last 24 hours of clean data
- ☐ **B)** Raw, unprocessed data as it arrived, often kept for a long time

- ☐ C) Only finished dashboard tables
- ☐ D) Data with all business rules already applied

Answer- B

MCQ 3.2 — Why organize Bronze data by date (e.g., year/month/day folders)?

- ☐ A) It's a legal requirement in every country
- ☐ B) It makes it easier to reprocess, audit, or re-run a specific day's data if something goes wrong
- ☐ C) It automatically removes duplicate records
- ☐ D) It makes the data unreadable to unauthorized users

Answer-B

MCQ 3.3 — Scenario: A retailer discovers 3 weeks later that a currency conversion bug corrupted last Tuesday's sales numbers in their dashboard. What allows them to fix this without asking customers to resend data?

- ☐ A) Nothing — the data is gone forever
- ☐ B) The raw Bronze data from that Tuesday is still there and can be reprocessed
- ☐ C) They must guess the correct numbers
- ☐ D) They restart the entire warehouse

Answer- B

MCQ 3.4 — Scenario: A compliance officer asks, "Can you show me exactly what data we received from the payment provider on March 5th, before any of our own processing touched it?" Which zone answers this?

- ☐ A) Gold
- ☐ B) Bronze
- ☐ C) A dashboard
- ☐ D) None — this information isn't kept anywhere

Answer – B

MCQ 3.5 — Which of these is a valid reason NOT to skip the Bronze zone and jump straight to cleaning data on arrival?

- ☐ A) Skipping Bronze saves no time at all
- ☐ B) Without Bronze, you lose the ability to redo cleaning/transformation later if requirements or bugs are found

- ☐ C) Bronze zones are required by every programming language
- ☐ D) There is no real reason, skipping it is always fine

Answer - B

Lab 4 — The Silver (Cleaned) Zone

Business Scenario

BrightCart's engineering team takes the raw Bronze order data and removes duplicates, fixes data types (e.g., "10" the text becomes 10 the number), and standardizes column names. This cleaned version is stored separately as "Silver," and multiple teams (Finance, Marketing, Data Science) all reuse it instead of re-cleaning the raw data themselves.

Think About It

- Why is it more efficient for multiple teams to share one Silver dataset instead of each team cleaning the raw data on their own?
- Is Silver data usually the final dataset shown on a dashboard? Why or why not?

A - Because silver layer provides a consistent intermediate datasets for all the team and it reduces repeated parsing work from each team making the process faster and saving time. And each team write their own cleaning logic to validate which may results in dieffrent outputs so providing silver data gives an intermediate data for everyone

B – No silver data is not the final dataset shown on a dashboard it's the Gold/curated data silver data is just a parsed and validated data

Multiple Choice Questions

MCQ 4.1 — Compared to raw Bronze data, Silver data is:

- ☐ A) Exactly identical, just renamed
- ☐ B) Parsed, validated, and deduplicated — a reusable, trustworthy intermediate version
- ☐ C) Only used for storing images
- ☐ D) Always deleted after 24 hours

Answer- B

MCQ 4.2 — Silver-zone data is best described as:

- ☐ A) The final dataset business executives view directly on dashboards
- ☐ B) A clean, validated dataset — still an intermediate step before final business-ready tables
- ☐ C) The same thing as the Bronze zone

☐ **D)** Data that only data scientists are allowed to see

Answer – B

MCQ 4.3 — Scenario: Three different teams keep writing their own scripts to clean the same messy raw order data, each producing slightly different results. What architectural fix solves this?

☐ **A)** Delete the raw data

☐ **B)** Build one shared Silver dataset that everyone reuses

☐ **C)** Give up on cleaning entirely

☐ **D)** Only let one team ever look at the data

Answer – B

MCQ 4.4 — Scenario: During cleaning, the Silver process removes exact duplicate order records that appeared twice due to a network retry. This step is an example of:

☐ **A)** Data aggregation for reporting

☐ **B)** Data validation/deduplication, a typical Silver-zone task

☐ **C)** Something that should only happen in Gold

☐ **D)** Something that should only happen in Bronze

Answer- B

MCQ 4.5 — Which of these tasks would you expect to happen in Silver, NOT in Bronze?

☐ **A)** Storing the file exactly as it arrived

☐ **B)** Converting a text field like "20.5" into an actual numeric value

☐ **C)** Keeping every historical raw file forever

☐ **D)** Ingesting data with zero changes

Answer – B

Lab 5 — The Gold (Curated) Zone

Business Scenario

BrightCart's Finance team needs a single, ready-to-use table: total sales per region per day, refreshed every morning, that plugs directly into their dashboard tool.

Think About It

- Why would the company build a separate "Gold" table instead of having Finance query the raw Bronze data directly every morning?
- What business value does a Gold table provide that Silver doesn't?

A – Because the data is raw you can not have it in your dashboards directly as it will be for business use and it can not give you an optimized data, having Gold data will give an business level optimized data for reporting and dashboards which will be a trustworthy, high quality datasets.

B – Silver data is just a parsed and validated data it is not for business while Gold data is ready to use for business high quality, trustworthy data which will be having proper star schema or aggregates which a business is needed.

Multiple Choice Questions

MCQ 5.1 — Gold-zone tables are typically:

- ☐ A) Raw and unprocessed
- ☐ B) Curated, aggregated, and business-ready — optimized for dashboards and reporting
- ☐ C) Only readable by engineers
- ☐ D) Rebuilt from scratch by hand every day

Answer – B

MCQ 5.2 — The simplest reason a BI dashboard tool (like Power BI, Tableau, or Excel) connects to Gold tables instead of raw Bronze files is:

- ☐ A) Gold data is already clean, aggregated, and in a consistent format ready for reporting
- ☐ B) Bronze files are usually password protected
- ☐ C) Gold tables are always smaller in storage size
- ☐ D) There's no real reason, it's arbitrary

Answer- A

MCQ 5.3 — Scenario: The CEO asks for "total revenue by region, updated daily" on a live dashboard. Which zone should power that dashboard?

- ☐ A) Bronze
- ☐ B) Silver
- ☐ C) Gold
- ☐ D) None — recompute from scratch each time from source systems

Answer- Gold

MCQ 5.4 — Scenario: A retail chain builds a Gold table called "fact_daily_store_sales" that joins cleaned sales data with store location and product category info, pre-aggregated by day. This is a good example of Gold because it is:

- ☐ A) Raw and untouched
- ☐ B) Business-friendly, pre-joined, and ready for fast reporting without extra work
- ☐ C) Something only engineers would ever look at
- ☐ D) A duplicate copy of Bronze

Answer- B

MCQ 5.5 — Which statement about Gold tables is FALSE?

- ☐ A) They are usually refreshed on a schedule (e.g., daily/hourly)
- ☐ B) They are meant to be queried directly by business users and dashboards
- ☐ C) They should always be rebuilt entirely from scratch by hand every time someone needs a number
- ☐ D) They are typically smaller and more aggregated than Silver

Answer – C

Lab 6 — Storage-First & Separating Compute from Storage

Business Scenario

BrightCart stores all its Bronze/Silver/Gold data in one central cloud storage location. Three different tools access this same data: the Finance team's SQL reporting tool, the Data Science team's Python notebooks, and an external auditor's reporting software.

Think About It

- Why can all three tools read the same data without needing their own separate copies or stepping on each other?
 - What would be the downside if data could only be accessed through one specific tool/database engine?
- A- Because they use three different tools they don't need separate copies for each other which may result in inconsistent data since its write once read many all can access the source a once without depending on each other and it reduces the storage cost by decoupling the storage from compute
- B- Then each team id dependent on one another which results in lower performance and they need separate copies for each which increases the storage cost and risk of inconsistent data.

Multiple Choice Questions

MCQ 6.1 — Separating "storage" (where data lives) from "compute" (the tools that process it) mainly allows:

- ☐ A) Data to disappear automatically after use
- ☐ B) Multiple tools/teams to access the same data independently, and scale processing power only when needed
- ☐ C) Guaranteed zero-latency access at all times
- ☐ D) Elimination of any need for security

Answer-B

MCQ 6.2 — Scenario: BrightCart's Marketing team has a huge one-time analysis to run this week, needing 10x more processing power than usual, but only for 3 days. With compute/storage separation, what's the sensible approach?

- ☐ **A)** Buy a new permanent server for 10x capacity
- ☐ **B)** Temporarily scale up compute power for those 3 days, then scale back down — storage stays untouched
- ☐ **C)** Copy all the data to a new physical location
- ☐ **D)** Refuse the request — it's not technically possible

Answer- B

MCQ 6.3 — In a traditional tightly-coupled database (compute and storage bundled together), what typically happens if you need more query power?

- ☐ **A)** Nothing changes, it scales itself infinitely for free
- ☐ **B)** You often have to scale up the entire machine (including storage you didn't need more of), which is costly and less flexible
- ☐ **C)** You simply add more storage disks and query speed increases automatically
- ☐ **D)** It's impossible to add more power at all

Answer- B

MCQ 6.4 — Scenario: BrightCart shuts down its Python analysis jobs for the weekend, but the SQL reporting tool still needs 24/7 access to the same Gold tables. Storage-first architecture allows this because:

- ☐ **A)** The data itself doesn't depend on any single compute engine being on
- ☐ **B)** Data disappears when compute is off
- ☐ **C)** Only one tool can ever exist at a time
- ☐ **D)** SQL tools cannot read shared storage

Answer- A

Lab 7 — Metadata & Data Catalogs

Business Scenario

A new analyst joins BrightCart and asks: "Where is the sales data? Who owns it? What does each column actually mean? Has it changed recently?" Without documentation, they'd have to ask around or dig through files manually.

Think About It

- What problems arise in a company if there's no catalog or documentation describing available datasets?
- How does a catalog help with trust and governance, not just findability?

A – If there is no catalog or documentation about datasets then people will not have information on who owns the sales, what is the sales data what does this column means and they have to manually find it out and for a large data set it takes so much time with catalog information they can easily find out all these information without wasting much effort and time

B – Catalog will help to clear definitions by identify what each column name means other wise different team understands it differently and governance gives permissions and row level security where only specified team can access specified details in a schema

Multiple Choice Questions

MCQ 7.1 — The main purpose of a data catalog is to:

- ☐ **A)** Physically store the data files
- ☐ **B)** Help people discover what data exists, who owns it, what it means, and where it came from
- ☐ **C)** Replace the need for any data storage

☐ **D)** Make data load faster over the internet

Answer – B

MCQ 7.2 — Scenario: An auditor needs to know exactly when a "revenue" column's calculation logic changed last year. A good catalog/metadata system should be able to show:

☐ **A)** Nothing — this isn't possible

☐ **B)** The history/lineage of changes made to that dataset over time

☐ **C)** Only the current value, never past changes

☐ **D)** The auditor's own notes, not the system's

Answer- B

MCQ 7.3 — Which of these is NOT typically part of a data catalog entry?

☐ **A)** Table owner

☐ **B)** Column descriptions

☐ **C)** The literal raw bytes of every file, duplicated

☐ **D)** Data lineage (where it came from, what transformed it)

Answer – C

MCQ 7.4 — Scenario: Two teams both have a table called "customers" with different meanings — one is "signed-up users," the other is "paying customers only." Without a catalog, what's the likely business risk?

☐ **A)** None, the names being the same causes no confusion

☐ **B)** Reports or decisions could be based on the wrong definition of "customers," leading to incorrect conclusions

☐ **C)** The database would refuse to run at all

☐ **D)** Both tables would automatically merge correctly

Answer- B

MCQ 7.5 — A well-maintained data catalog mainly builds:

☐ **A)** Faster internet speeds

☐ **B)** Trust and shared understanding of data across teams

☐ **C)** More raw storage space

☐ **D)** Automatic dashboards with no design needed

Answer - B

Capstone Scenario — Design the Full Pipeline (No Coding)

Business Scenario

"BrightCart" is launching a new mobile app and wants to track user activity (clicks, screen views, purchases) end-to-end — from raw ingestion to an executive dashboard.

Task

On paper (or in this document), sketch your answer to each:

1. Where would raw app event data land first, and why (name the zone)?
2. What cleaning/validation steps would move it to the next zone?
3. What final business-ready table would you build, and who would use it?
4. If the marketing team later finds a bug in how "purchase" events were being counted, which zone would you fix and re-run from — and why does keeping that earlier zone matter?
5. Would you use a data lake, warehouse, or lakehouse approach overall for this app? Justify your choice in 2–3 sentences.

1. Ingestion

The raw event data from app first land in Bronze zone which because it is a ingestion of raw data it goes to raw/bronze zone first

2. Processing/Transform

Next Zone is Silver Zone where deduplication of data will be done and the data will be parsed like timestam parcing and missing/unused rows will be dropped and strings will be converted into numbers and validation of mandetory fileds by checking null values so on..

3. Serving/Analysis

The Final business ready table would be a Gold Zone table which has high quality analytical ready data. It will be used by Analytics for reporting business owners for Dashboards.

4. If the bug is found later in a cleaned data then we can re run from bronze zone and fix the silver zone again. If Bronze zone has not been kept then we don't be having a raw data where we can fall back and re run to when a bug appears so keeping Bronze zone helps in reprocessing if a bug appears.
5. I would suggest to use Data Lakehouse for this because if there is any structured or semi structured or unstructured raw data then lakehouse will divide that raw data accordingly By creating a separate layers which makes things easy and also it avoids data swamping problem and also provide low cost storage like lake also it adds transactional metadata and query optimization making analytics more robust than a lake.

DATA PIPELINE OF BRIGHTCART MOBILE APP

Ingestion(Ingestion of Raw data from app)(Nifi,Flume)->

storage(Raw/Bronze zone)(Storing the ingested raw data from app in Bronze Zone)(DataLakehouse like OneLake or DeltaLake)->

Processing(Silver zone)(Processing is validating and parsing the raw data in silver zone)(MapReduce,spark,Flink)->

Serving/Analysis(Gold Zone)(publishing the Business ready high quality data to dashboards, reportings, analysis)(Power BI, Tableau)->

Monitoring(Monitoring the pipeline health)

